INTRODUCTION

This document explains the design and implementation of a basic interactive Graphical User Interface (GUI) application built with Java Swing. Swing is a toolkit that ships with Java and lets programmers build desktop windows made of components such as frames, panels, labels, and buttons. The application displays a welcome message inside a window and lets the user change the message text and the background colour by clicking buttons.

The project demonstrates the use of JFrame, JPanel, JLabel and ImageIcon, together with two core object-oriented programming ideas: inheritance and constructors.

OBJECTIVE

- Create a window (JFrame) titled "Java Swing Assignment" with a size of 800x800 pixels.
- Add a JPanel to the frame and a JLabel to the panel.
- Apply inheritance by extending the JFrame class, and use constructors to set up the objects.
- Use ImageIcon to set a personal application icon.
- Make sure the application closes when the close button is clicked.
- Add extra functionality to change the label text and the panel background colour.

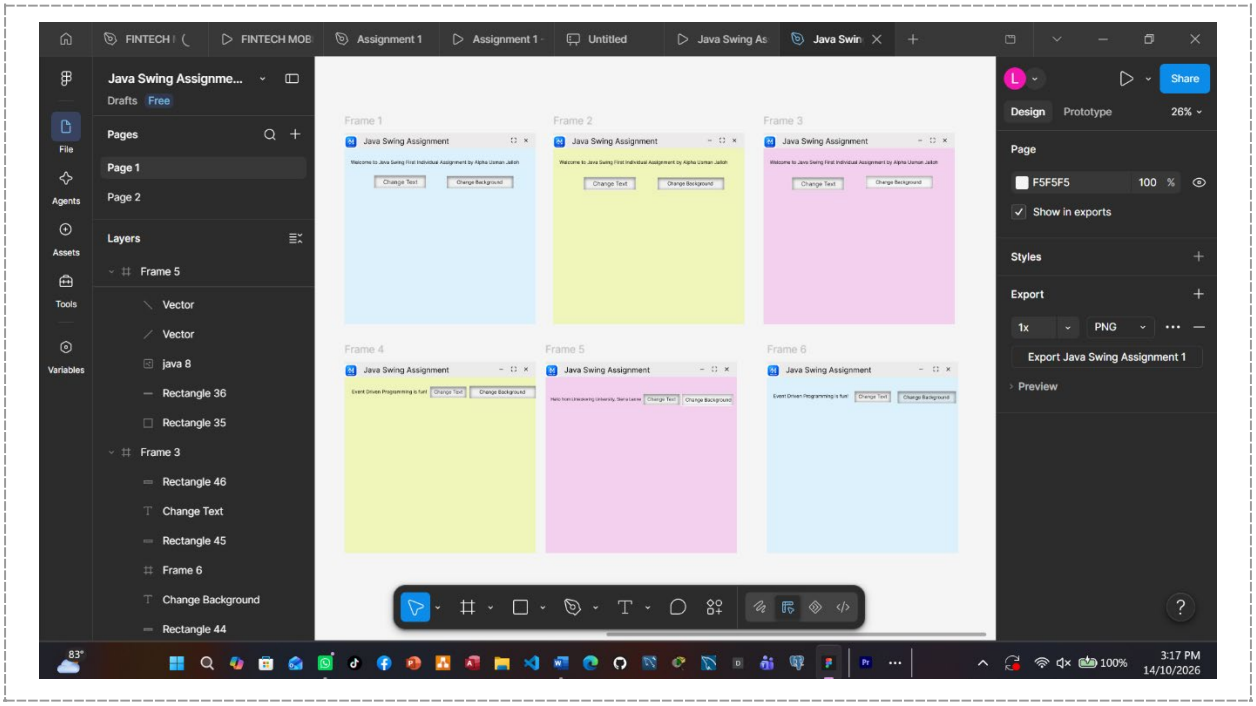
TOOLS USED

Tool	Purpose
Java Development Kit (JDK)	Compiling and running the Java program
IntelliJ IDEA / VS Code	Writing and testing the source code
Figma	Designing the wireframe of the interface
Git and GitHub	Version control and submission of all project files

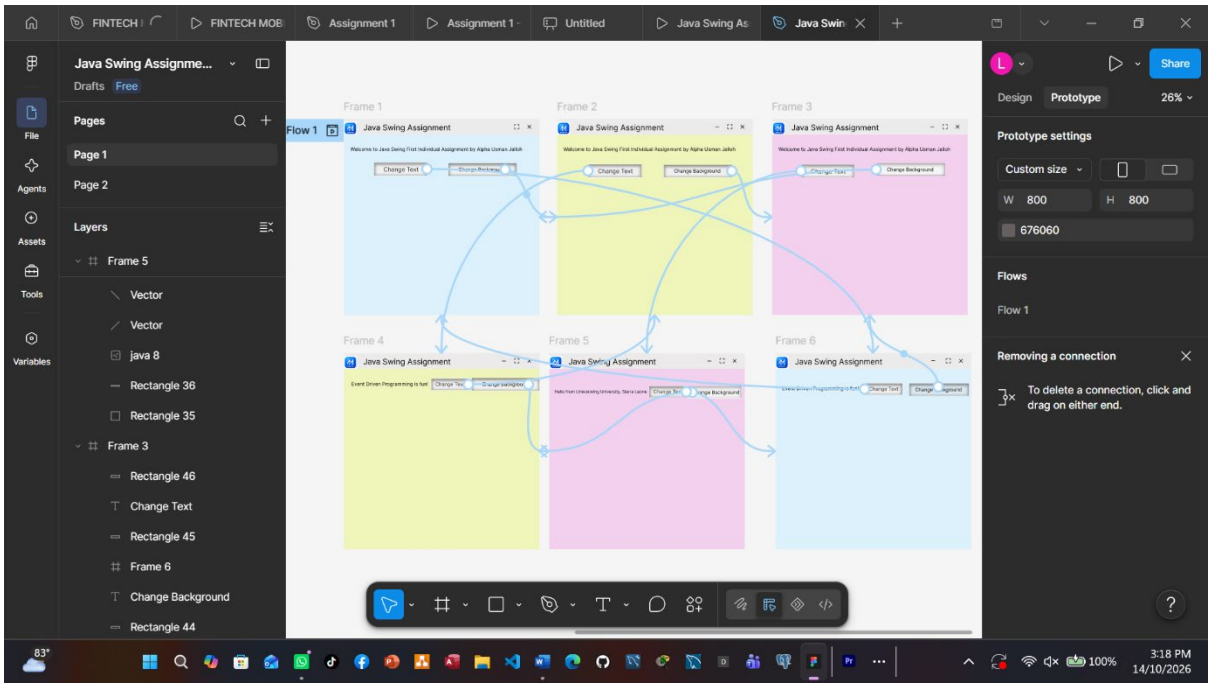
WIREFRAME

A wireframe is a simple blueprint of a screen that shows where each element will be placed before any code is written. The wireframe was created in Figma using an 800x800 pixel frame. It contains the title bar (Java Swing Assignment), a panel filling the window, the welcome label at the top of the panel, and two buttons placed below the label: "Change Text" and "Change Background".

SCREENSHOT TWO



SCREENSHOT TWO



HOW THE REQUIREMENTS WERE MET

Requirement	Implementation
Class called SwingAssignment	public class SwingAssignment is the main class of the program.
JFrame variable named frame	private JFrame frame; instantiated as new CustomFrame(...).
JPanel variable named panel	private JPanel panel; instantiated with new JPanel().
JLabel variable named label	private JLabel label; created with the required welcome text.
Add label to panel, panel to frame	panel.add(label); and frame.add(panel);
Title and 800x800 size	Passed to the CustomFrame constructor: "Java Swing Assignment", 800, 800.
Application closes on close button	setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE)
Own application icon and ImageIcon	new ImageIcon("icon.png") passed to setIconImage().
Inheritance and constructor	CustomFrame extends JFrame and has its own constructor.
Addition: change text and background	Two JButtons with ActionListeners update label text and panel colour.

KEYS CONCEPTS EXPLAINED

Inheritance. Inheritance lets one class reuse the features of another. The class CustomFrame uses the keyword extends to inherit from JFrame, so it automatically has all the abilities of a window (size, title, visibility) and we only add our own settings.

Constructor. A constructor is a special method with the same name as its class. It runs automatically when an object is created with new. In CustomFrame, the constructor sets the title, size, close behaviour, screen position and icon. In SwingAssignment, the constructor builds the entire interface.

ImageIcon. ImageIcon loads a picture from a file. Here it loads icon.png and the image is applied as the window icon using setIconImage(). A check on getIconWidth() ensures the program still runs if the file is missing.

Event handling. An event is an action such as a button click. An ActionListener is attached to each button; when the button is clicked, the code inside the listener runs. This is the central idea of event driven programming.

SOURCE CODE

The complete source code of SwingAssignment.java is shown below.

```
// Source code is decompiled from a .class file using FernFlower decompiler
(from Intellij IDEA).
import java.awt.Color;
import java.awt.Font;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JPanel;
import javax.swing.SwingUtilities;

public class SwingAssignment {
   private JFrame frame = new CustomFrame("Java Swing Assignment", 800, 800);
   private JPanel panel = new JPanel();
   private JLabel label;
   private final String[] messages = new String[]{"Welcome to Java Swing First
Individual Assignment by Alpha Usman Jalloh", "Event Driven Programming is
fun!", "Hello from Limkokwing University, Sierra Leone"};
   private final Color[] colours = new Color[]{new Color(224, 242, 254), new
Color(220, 252, 231), new Color(254, 243, 199), new Color(252, 231, 243)};
   private int messageIndex = 0;
   private int colourIndex = 0;

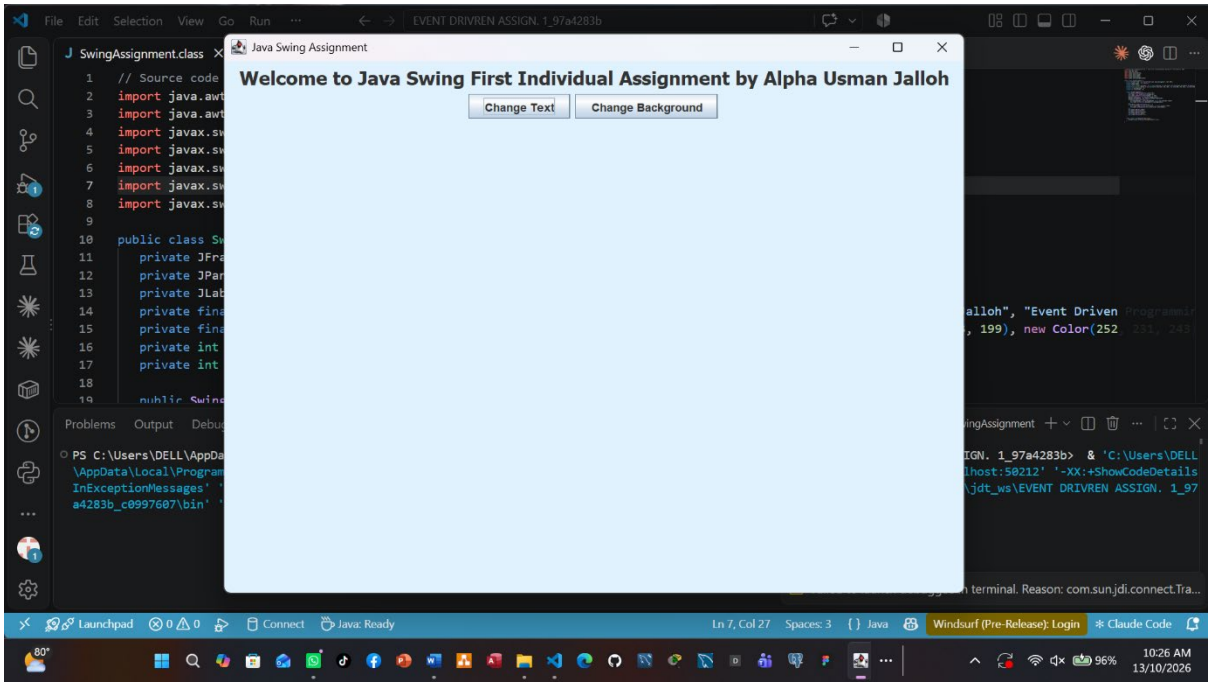
   public SwingAssignment() {
      this.panel.setBackground(this.colours[0]);
      this.label = new JLabel(this.messages[0]);
      this.label.setFont(new Font("Tahoma", 1, 20));
      JButton textButton = new JButton("Change Text");
      JButton colourButton = new JButton("Change Background");
      textButton.addActionListener((e) -> {
         this.messageIndex = (this.messageIndex + 1) % this.messages.length;
         this.label.setText(this.messages[this.messageIndex]);
      });
      colourButton.addActionListener((e) -> {
         this.colourIndex = (this.colourIndex + 1) % this.colours.length;
         this.panel.setBackground(this.colours[this.colourIndex]);
      });
      this.panel.add(this.label);
      this.panel.add(textButton);
      this.panel.add(colourButton);
      this.frame.add(this.panel);
      this.frame.setVisible(true);
   }

   public static void main(String[] args) {
      SwingUtilities.invokeLater(SwingAssignment::new);
```

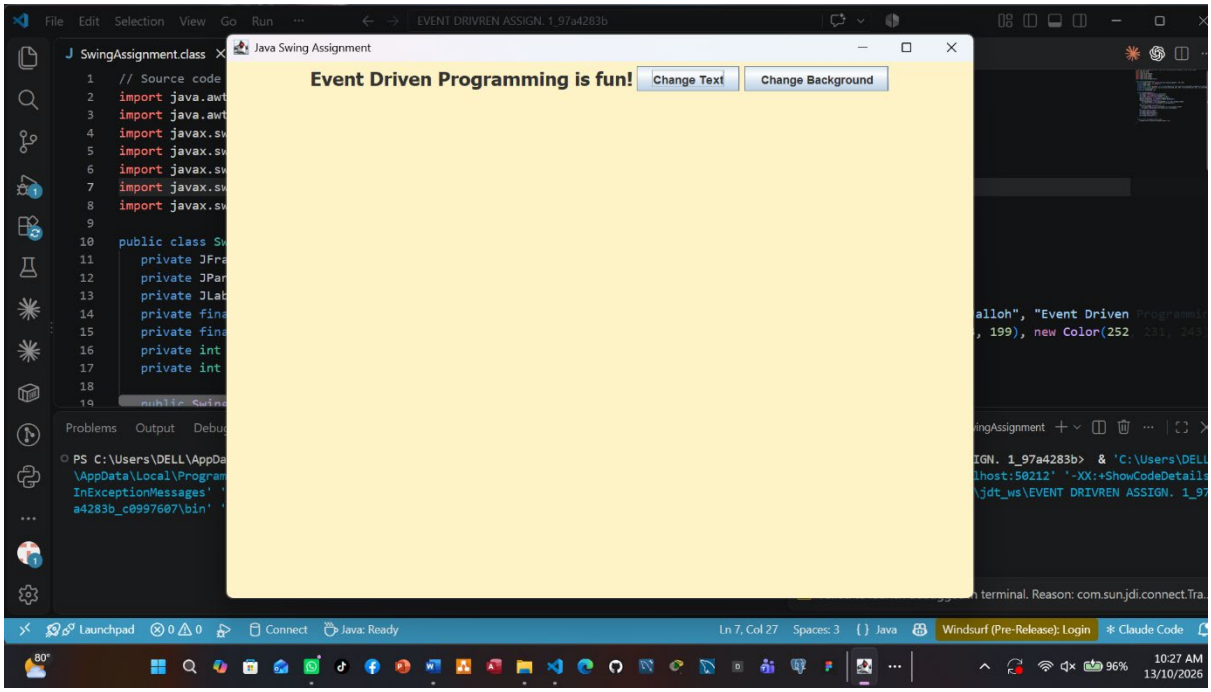
```
    }
}
```

OUTPUT SCREENSHOTS

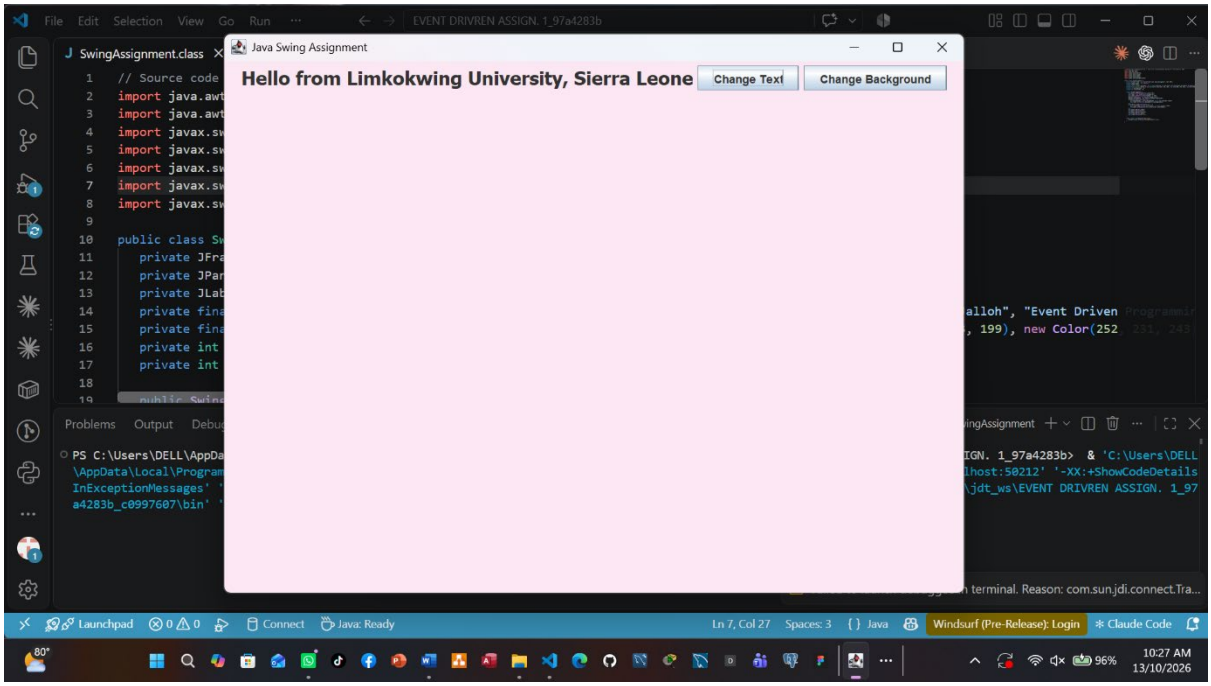
Screenshot 1:



Screenshot 2:



Screenshot 3:



CONCLUSION

The assignment produced a working Java Swing application that meets every requirement in the brief. It showed how a window is built from a JFrame, JPanel and JLabel, how inheritance and constructors organise code, how ImageIcon is used for an icon, and how event listeners make an application interactive.

GitHub repository: [https://github.com/Leeson001/BASIC-INTERACTION-GUI.git]

REFERENCE

1. Oracle (n.d.) The Java Tutorials: Creating a GUI with Swing. Available at: https://docs.oracle.com/javase/tutorial/uiswing/ B(Accessed: October 2026).